

Lecture 11

In Class Exercise

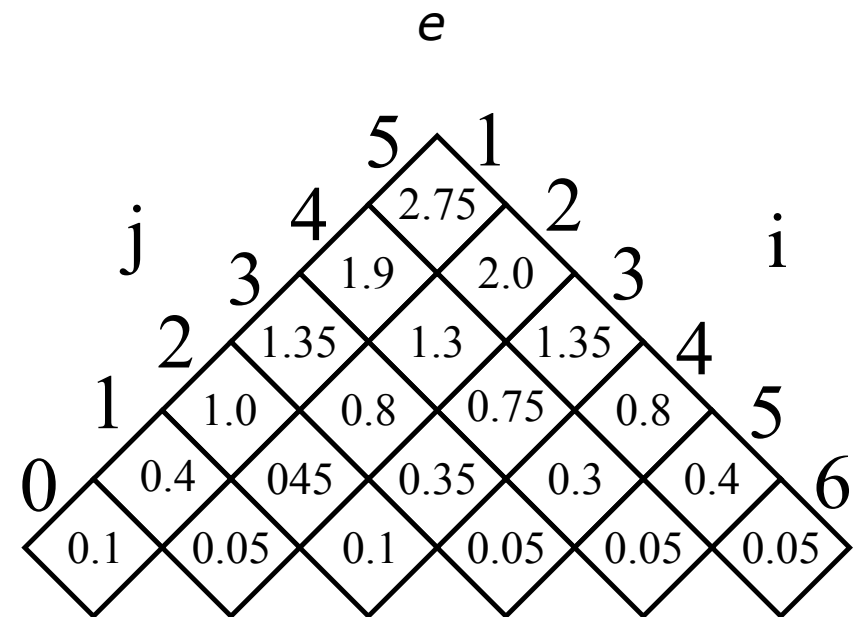
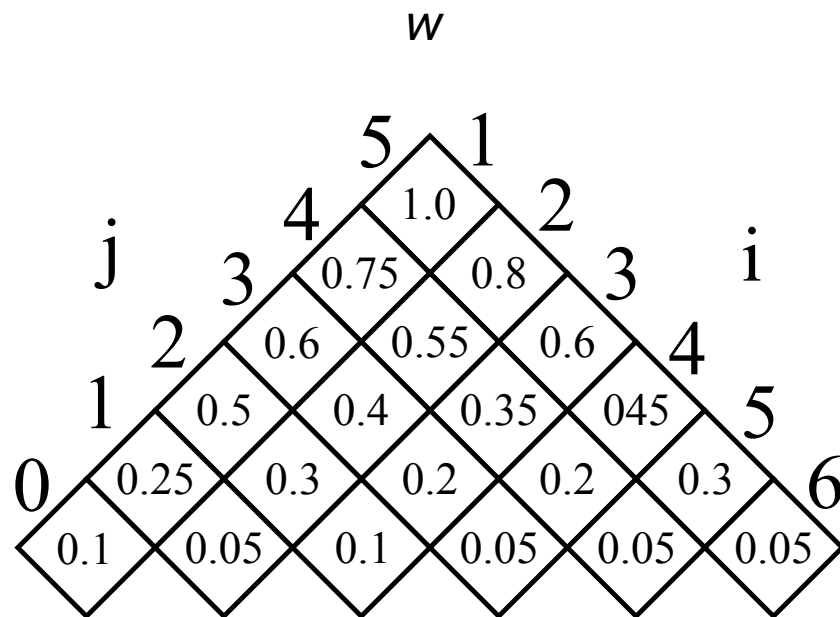


Exercise 1

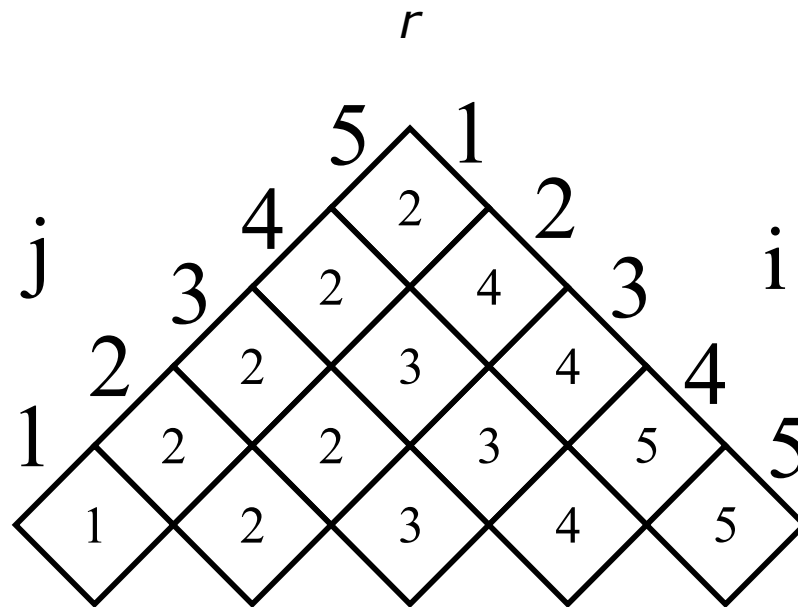
Determine the cost and the structure of an optimal binary search trees using the 5 keys provided below. Show the resulting w (sum of all probabilities), e (expected search cost), and r (root) tables. Finally, show the resulting optimal binary search tree.

i	0	1	2	3	4	5
p_i		0.1	0.15	0.05	0.1	0.2
q_i	0.1	0.05	0.1	0.05	0.05	0.05

Exercise 1 (solution)



Exercise 1 (solution)



Exercise 1 (solution)

Optimal Binary Search Tree

